The mean of a variable is defined as the sum of the observations divided by the number of observations. If the variable x assumes n values x_1, x_2 … x_n then the mean, , is given by summing of all x values divided by the number of x values.

$$\bar{X} = \frac{X_1+X_2+X_3+X_4+\cdots+X_n}{n} = \frac{1}{n}\sum_{i}^{n}=_1 X_i,$$

This formula is for ungrouped or raw data.

Example 1: Calculate the mean for given data 2,4,6,8,10.

Direct Method $\bar{X} = \frac{\sum X}{n}$

Where $\sum X$ = Sum of values

N = No. of items

Solution:

$$\bar{X} = \frac{2+4+6+8+10}{5} = \frac{30}{5} = 6$$

Short- cut method:

The formula for finding mean,

$$\bar{X} = A + \frac{\sum d}{n}$$

Where A= the assumed mean or any value of X

d = deviations of each value from assumed mean.

Example 2: A student's marks in 5 subjects are 75,68,80,92,and 56. Find his average mark.

Solution:

X	d = X - A
75	7
68 → A	**0**
80	12
92	24
56	-12
Total	**31**

$$\bar{X} = A + \frac{\sum d}{n}$$

$$= 68 + \frac{31}{5}$$

$$= 68 + 6.2$$

$$= 74.2$$

12.11 Standard Deviation (σ)

The measures of central tendency serve to locate the center of the distribution, but they do not reveal how the items are spread out on either side of the center. This characteristic of a frequency distribution is commonly referred to as dispersion. The degree of variation is evaluated by various measures of dispersion. There are two kinds of measures of dispersion, namely

1. Absolute measure of dispersion
2. Relative measure of dispersion

Absolute measure of dispersion indicates the amount of variation in a set of values in terms of units of observations.

Relative measures of dispersion are free from the units of measurements of the observations. They are pure numbers. They are used to compare the variation in two or more sets, which are having different units of measurements of observations.

Standard Deviation is one of the methods of Absolute measure of dispersion. Karl Pearson introduced the concept of standard deviation in 1893. Standard deviation is also called Root-Mean Square Deviation. The reason is

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